Fully automatic volume segmentation of infrarenal abdominal aortic aneurysm computed tomography images with deep learning approaches versus physician controlled manual segmentation

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ABSTRACT

Objective: Imaging software has become critical tools in the diagnosis and decision making for the treatment of abdominal aortic aneurysms (AAA). However, the interobserver reproducibility of the maximum cross-section diameter is poor. This study aimed to present and assess the quality of a new fully automated software (PRAEVAorta) that enables fast and robust detection of the aortic lumen and the infrarenal AAA characteristics including the presence of thrombus.

Methods: To evaluate the segmentation obtained with this new software, we performed a quantitative comparison with the results obtained from a semiautomatic segmentation manually corrected by a senior and a junior surgeon on a dataset of 100 preoperative computed tomography angiographies from patients with infrarenal AAAs (13,465 slices). The Dice similarity coefficient (DSC), Jaccard index, sensitivity, specificity, volumetric similarity (VS), Hausdorff distance, maximum aortic transverse diameter, and the duration of segmentation were calculated between the two methods and, for the semiautomatic software, also between the two observers.

Results: The analyses demonstrated an excellent correlation of the volumes, surfaces, and diameters measured with the fully automatic and manually corrected segmentation methods, with a Pearson's coefficient correlation of greater than 0.90 ($P < .0001$). Overall, a comparison between the fully automatic and manually corrected segmentation method by the senior surgeon revealed a mean Dice similarity coefficient of 0.95 $\pm$ 0.01, a Jaccard index of 0.91 $\pm$ 0.02, sensitivity of 0.94 $\pm$ 0.02, specificity of 0.97 $\pm$ 0.01, VS of 0.98 $\pm$ 0.01, and mean Hausdorff distance per slice of 4.61 $\pm$ 7.26 mm. The mean VS reached 0.95 $\pm$ 0.04 for the lumen and 0.91 $\pm$ 0.07 for the thrombus. For the fully automatic method, the segmentation time varied from 27 seconds to 4 minutes per patient vs 5 minutes to 80 minutes for the manually corrected methods ($P < .0001$).

Conclusions: By enabling a fast and fully automated detailed analysis of the anatomic characteristics of infrarenal AAAs, this software could have strong applications in daily clinical practice and clinical research. (J Vasc Surg 2021;74:246-56.)

Keywords: Abdominal aortic aneurysm; Endovascular aortic repair; Artificial intelligence; Automatic segmentation; Volume; Deep learning; Thrombus

The immediate decision about the size at which an aneurysm should be repaired is framed by the balance between the risk of aneurysm rupture (which is still fatal in >80% cases)[1] and the risk of operative mortality. Currently, the evidence for the threshold for abdominal aortic aneurysm (AAA) repair is based on aortic diameter, not volume measurements. In general, the maximum cross-section diameter (outer to outer wall) is considered to be related to the risk of rupture owing to increased wall tension with increased radial growth. The cumulative yearly rupture rate is of 3.5% in AAAs of 5.5 to 6.0 cm in diameter, 4.1% in AAAs of 6.1 to 7.0 cm, and 6.3% in AAAs greater than 7 cm.[2]

Various imaging modalities have been studied for visualization of AAAs, such as ultrasound examination, computed tomography angiography (CTA), and magnetic resonance imaging. As of today, CTA with a slice thickness of 1 mm or less plays a key role in assessing the extent of disease and therapeutic decision making and planning.[3] CTA is also the currently recommended imaging modality for the diagnosis of rupture and is an important tool during follow-up after repair,[4] enabling analysis in three perpendicular planes, construction of a centerline, and accurate diameter measurement with dedicated postprocessing software. However, the interobserver reproducibility of maximum cross-section diameter is poor, with 87% comparisons being outside the clinically accepted range of $\pm$5 mm of diameter measurement variation.[5] Much of the difference is probably attributable to inadequate reporting standards concerning the specification of the aortic axis, plane of measurement, and caliper placement. This variability is

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Author conflict of interest: none.

All data generated or analyzed during this study are included in this published article.

Additional material for this article may be found online at www.jvascsurg.org.

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The editors and reviewers of this article have no relevant financial relationships to disclose per the JVS policy that requires reviewers to decline review of any manuscript for which they may have a conflict of interest.

0741-5214

https://doi.org/10.1016/j.jvs.2020.11.036

of particularly high clinical significance, because the number of patients considered for AAA repair, based on a diameter threshold, may vary from 5% to 24%, depending on the radiologists.[5]

Aneurysm volume has been previously proven to be more accurate than maximal aneurysm diameter in evaluating the aneurysmal sac and risk of rupture.[6] However, the automatic segmentation of the AAA is challenging owing to the heterogeneity of the AAA morphology, size, curvature, composition, and the close localization with the surrounding tissues.[7] Hence, most of the proposed algorithms require user interaction and/or prior lumen segmentation along with centerline extraction. These semiautomatic methods remain tedious and time consuming because it is done in the slice-to-slice process.[8]

In this article, we present an innovative software implementing artificial intelligence with dedicated algorithms to provide fully automatic segmentation, independent of human manual appreciation. This new software allows for the automatic detection of the main characteristics of AAAs (including an inner section [the lumen] and an outer fatty part [the thrombus]), which could be easily translated into clinical practice, to alleviate the burden of repetitive tasks, reduce analysis time, and improve reproducibility in comparing the diameters and volumes of AAAs on CTA.

ARTICLE HIGHLIGHTS

- **Type of Research:** Single-center, observational retrospective analysis
- **Key Findings:** Quantitative comparison between fully and semiautomatic segmentation over 100 computed tomography angiographies of abdominal aortic aneurysms revealed robust detection of the aortic lumen (volume similarity, 0.95 $\pm$ 0.04), thrombus (volume similarity, 0.91 $\pm$ 0.07), global volume (volume similarity, 0.98 $\pm$ 0.01), and diameters with the fully automatic method, and a time gain of 95%.
- **Take Home Message:** Automatic volume segmentation by PRAEVAorta could have strong applications in daily clinical practice and clinical research.

METHODS

Dataset. Preoperative CTA images were obtained from 100 consecutive patients with an infrarenal AAA who had multidetector CT scanners with 1-mm-thick slices of the entire abdominal aorta and arterial phase intravenous injection of contrast liquid at Bordeaux University Hospital (Bordeaux, France). All methods were carried out following the French Regulatory Health Authorities. Images were given in Digital Imaging and Communications in Medicine format providing a matrix of size 512 $\times$ 512 for each slice. Patients with pararenal or thoracoabdominal aneurysms, associated iliac artery aneurysms, ruptured aneurysms, aortic dissections, and noncontrast CTs were excluded from the analysis. The experimental protocol was approved by the institutional review board, and all patients gave informed consent. The research was conducted following the principles outlined in the Declaration of Helsinki.

Fully automatic segmentation methodology. Fully automatic segmentation was provided by PRAEVAorta, an automatic segmentation software developed by Nurea (https://www.nurea-soft.com/). This software is only available for research for now, but should be commercially available in the near future. The method used for aneurysm segmentation from CT-scan images consists of four sequential steps: image preprocessing, segmentation of the aortic lumen, and thrombus. The boundary between the lumen and the thrombus is considered as the inner wall and the boundary between the thrombus and the background tissues as the outer wall. A machine learning approach in AAA imaging analysis was used[9] and 38 CTAs of infrarenal AAAs (different from the 100 CTAs analyzed in this article) were used for training the algorithms. Data augmentation was performed to reconstruct artificial realistic images and corresponding segmentations and to decrease the number of training original CTAs without losing performances.

Models were created using Keras and Tensorflow framework. The models designed by the PRAEVAorta software are based on convolutional neural networks similar to the classical Unet network. Both training and testing were done on one Intel Core i7 8750H 2.2GHz with 32 GB of RAM equipped with a Nvidia RTX 2080 Max-Q, under Windows 10 Pro 64 bits for training part and on Linux Ubuntu 18.04.4 LTS 64 bits for prediction part.

Image preprocessing. Preprocessing, which is a mandatory step before image segmentation, was fully automated in the PRAEVAorta software. Standard image filtering and denoising algorithms for medical imaging were used. Based on pixel characteristics (black pixels around the body, scan table detection), the first region of interest is extracted, roughly centered around the aorta to decrease computing time.

Localization of anatomical landmarks. As the first step, some anatomical landmarks were located to initialize the aorta segmentation and to crop the original volume. The first caliper was placed on the lowest part of the lowest renal artery and the second caliper at the aortic bifurcation.

Lumen segmentation. The method for lumen segmentation was designed to allow automatic detection and localization of the aorta. The method was built to discriminate contrasted arterial systems from surrounding tissues, which can be carried out with simple thresholding of the pixels and voxels in a predefined area. Based on the Hounsfield unit (HU), in contrast-enhanced CTAs, the lumen is expected to have intensities between 200 and 400. However, spine or bones can have similar pixel intensities and can be located very close to the aorta, making simple thresholding difficult to automate. By mixing statistical analyses and specific machine learning techniques, anatomical structures are correctly isolated in a fast and efficient way. Then, the inner wall segmentation was carried out as the boundary of the lumen region.

Thrombus segmentation. After the lumen region was selected and separately identified from the rest of the aneurysm, it was set as the initial level for segmenting the thrombus region based on the same principle of pixel and voxel identification. A second algorithm is used to specifically identify the lumen-surrounding tissues and correctly discriminate the walls of the aorta from the vena cava and the duodenum, which can have a similar texture and pixel intensities. Morphologic operations (closing, smoothing) are then performed to ensure realistic arteries' shapes (no holes or angles). By merging lumen segmentation and wall segmentation with morphologic operations (closing and opening), the algorithm fully reconstructs the arteries with accurate discrimination of lumen and wall.

Visualization and maximum transverse diameter computation. The two-dimensional visualization was performed using a pixel matrix with a gray-scale color map, showing CTA slices in z-direction one by one and the contours of the segmented regions, so that they could be compared with manually drawn contours by experts. Metrics were then applied to benchmark the proposed methods quantitatively. The three-dimensional visualization was also used to render, in real time, the arterial system, visualize the centerline, and the difference between lumen and wall.

To compute the quantitative measures representing aorta morphology, the aorta centerline and oblique cross-sectional planes normal to the centerline were extracted. The centerline was computed based on the lumen segmentation and defined as the path along the lumen keeping the largest distance to lumen boundaries. The maximum transverse diameter was then defined as the maximum diameter computed in the planes orthogonal to the centerline, including the lumen and the thrombus. Branches of the arterial system are discriminated based on the centerline to allow specific volume analysis and distances between renal and iliac arteries.

Validation methodology. For each patient, manual semiautomatic segmentation of the aortic lumen and the thrombus was independently performed by one senior expert vascular surgeon and one junior trainee in vascular surgery using a contour drawing tool within the image processing software itk-SNAP (v 3.8.0). The choice of a senior and a junior operator was supported by the search for possible interoperator variability by considering two operators of strictly distinct levels. The expert surgeon was a national and international expert ensuring the sizing of all the endovascular aortic repairs (EVAR), fenestrated EVAR and branched EVAR of a large French university center. The junior surgeon was a resident specializing in vascular and endovascular surgery, involved in aortic sizing and endograft implantation procedures. In our country and our center, the majority of preprocedure analyses and measurements are carried out by a surgical team. Thus, the choice of the comparison between the PRAEVAorta software and a surgical analysis, and not a radiologic one, was guided by a desire for therapeutic analysis, which is more complete and more precise than a diagnostic analysis.

Quantitative validation of the algorithm was performed using the same CT scan image datasets. The results of the fully automatic segmentation were then compared with the manually corrected method using the following distinct metrics: overlap-based, volume-based, and surfaces-based metrics.

Overlap-based metrics.

- The Dice similarity coefficient (DSC) of the segmented volumes,[10] and
- the Jaccard index (JAC), defined as:

$$\mathrm{JAC} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$$

where TP is the true positive, FP the false positive (periaortic tissues, such as the duodenum or the vena cava, taken into account in the segmentation), and FN the false negative (aortic segments not taken into account in segmentation). JAC is smaller than DSC except at the extrema {0, 1} where they are equal.

The two metrics are related according to the formula:

$$\mathrm{JAC} = \mathrm{DSC}/(2-\mathrm{DSC})$$

- The true positive rate, or sensitivity, measures the portion of positive voxels in the ground truth that is also identified as positive by the segmentation being evaluated[10]:

$$\mathrm{Sensitivity} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$$

- The true negative rate, or specificity, measures the portion of negative voxels (background) in the ground truth segmentation that is also identified as negative by the segmentation being evaluated[10]:

$$\mathrm{Specificity} = \mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$$

where TN is the true negative.

Volume-based metric.
The volumetric similarity (VS) is a measure that considers the volumes of the segmented areas. It represents the absolute volume difference divided by the sum of the compared volumes[10]:

$$VS = 1 - abs(FN - FP)/(2 - TP + FP + FN)$$

Surface-based metric. The Hausdorff distance (HD) is the maximum distance of a segmented volume to the nearest point in the other segmented volume.[11]

$$HD(A,\ B) = max(h(A,\ B),\ h(B,\ A))$$

where h(A, B) is called the directed HD and given by the formula:

$$h(A, B) = \max_{a \in A} \min_{b \in B} |a - b|$$

where |a − b | is the Euclidean distance.

Maximum aortic transverse diameter. In each original slice of the image, the aortic transverse diameter is computed as the diameter of the smallest circle containing the pixels identified as part of the aorta in the plane normal to the centerline. The maximum diameter is then identified.

Computational analysis time. The duration of the semiautomatic segmentation manually corrected by the senior surgeon and the junior surgeon was reported and compared with the fully automatic method.

Interobserver variability of manual segmentation. The variability among users was measured as the average of absolute interobserver differences.

RESULTS

To evaluate the results of the segmentation obtained with this new fully automatic software, we performed a quantitative comparison with the results obtained from a semiautomatic segmentation manually corrected by a senior and a junior surgeon on a dataset of 100 CTAs from patients with infrarenal AAAs. We analyzed and compared all the CTA slices available for each patient, from the lowest renal artery to the aortic bifurcation, that is, 13,465 slices. Representative images of the fully automatic and manually corrected segmentation of the aortic lumen and the intraluminal thrombus are presented in Fig 1. The analyses demonstrated an excellent correlation of the volumes and surfaces measured with the fully automatic and manually corrected segmentation methods, with a Pearson's coefficient correlation of greater than 0.90 ($P < .0001$;Fig 2).

The results on the metrics used to evaluate the segmentation errors are presented in Table I for the global volume, Supplementary Table I (online only) for the lumen, and Supplementary Table II (online only) for the thrombus. Overall, comparison between the fully automatic and manually corrected segmentation method by the senior surgeon revealed a mean DSC of 0.95 ± 0.01, JAC of 0.91 ± 0.02, sensitivity of 0.94 ± 0.02, specificity of 0.97 ± 0.01, VS of 0.98 ± 0.01, and mean HD/slice of 4.61 ± 7.26 mm (Fig 3). For the lumen, the same comparison revealed a mean DSC of 0.93 ± 0.05, JAC of 0.86 ± 0.07, sensitivity of 0.96 ± 0.07, specificity of 0.99 ± 0.02, and VS of 0.95 ± 0.04 (Supplementary Fig 1, online only), whereas for the thrombus, the DSC was 0.81 ± 0.10, JAC was 0.69 ± 0.13, sensitivity was 0.76 ± 0.11, specificity was 0.94 ± 0.02, and VS was 0.91 ± 0.07 (Supplementary Fig 2, online only).

The semiautomatic manual segmentations demonstrated low interobserver variability, with intraclass coefficient values of greater than 0.90, and good correlation of the volumes and surfaces, with a Pearson's coefficient correlation of greater than 0.90 ($P < .0001$) (Supplementary Fig 3, online only).

Over the 100 analyzed CTA scans, FP (pixels wrongly identified as thrombus or lumen) were predicted in only 3 of 100 slices (1/10 slices for the lumen, and 1.2/10.0 slices for the thrombus). These were related to the propagation to the vena cava, the duodenum, or the inferior mesenteric artery when they have similar intensity levels (Fig 4).

The mean time for automatic segmentation was only 90 seconds per patient (range, 27 seconds to 4 minutes) vs 35 minutes per patient for the manually corrected methods (range, 5 minutes to 80 minutes; $P < .0001$), for a time saving of 95% (Supplementary Fig 4, online only).

The analyses demonstrated an excellent correlation of the maximum aortic transverse diameters measured with the fully automatic and manually corrected segmentation methods (Table II), with a Pearson's coefficient correlation of 0.94 vs the senior surgeon ($P < .0001$; Fig 5) and a 2.3 ± 2.1 mm mean discrepancy between the two (median, 1.8; with a difference of ≥5 mm in 8%). The discrepancy was more important between the senior and the junior surgeon 2.8 ± 3.8 mm (median, 1.7 mm; with a difference of ≥5 mm in 24% between the junior and the senior surgeon, resulting in a 13% difference between the junior surgeon and the fully automatic method) with a lower Pearson's coefficient correlation (0.84). However, a paired one-way analysis of variance test showed no significant difference between the fully automatic method and the two surgeons ($P = .17$). Overall, the coefficient of variation of the absolute difference between the fully automatic and the manually

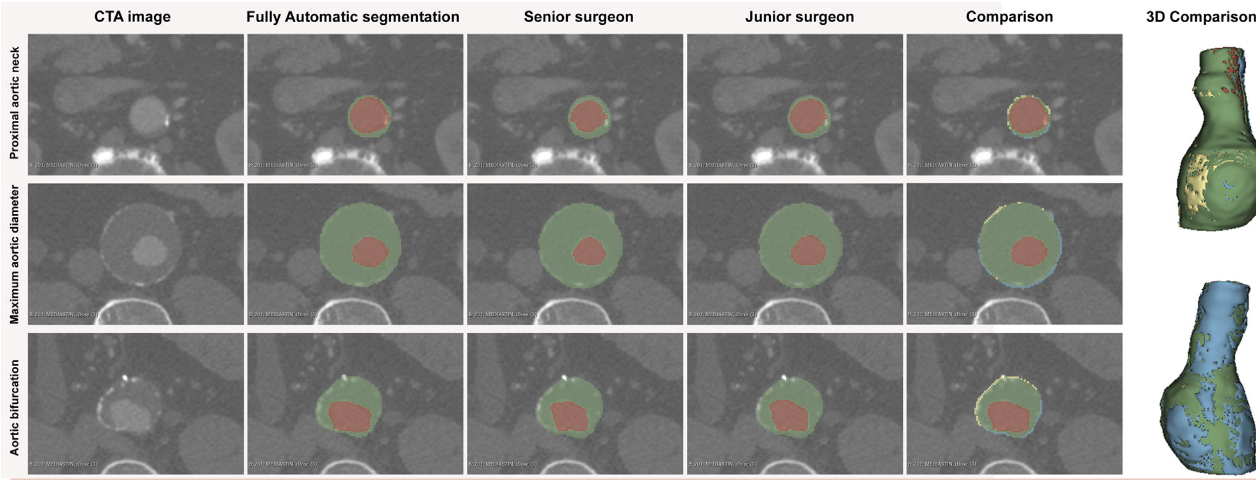


Fig 1. Representative images of the segmentation of the aortic lumen (*red*) and the intraluminal thrombus (*green*). Computed tomography angiography (*CTA*) cross-sections from a patient with infrarenal abdominal aortic aneurysm (AAA) are shown at three levels of the segmentation: the proximal neck, the maximum aortic diameter, and the aortic bifurcation. For each level, the fully automatic segmentation, the manually corrected segmentation by the senior surgeon and the junior surgeon. A comparison of the results obtained with the fully automatic segmentation and the manually corrected segmentation by the senior surgeon are also displayed, showing the lumen common to manual segmentation and automatic segmentation (*green*), the thrombus common to manual segmentation and automatic segmentation (*yellow*), the false negatives (*red*), and the false positives (*blue*).

corrected method (senior surgeon) was the highest for the surface measurements (113%), followed by the diameters (92%) and the volumes (76%).

DISCUSSION

Our results demonstrated the precision and robustness of these fully automatic detection algorithms in 100 CTAs of different infrarenal AAA anatomies, with a good correlation with the manually corrected segmentation performed by an expert senior surgeon and a junior surgeon. This approach is based on deep convolutional networks that can extract deep features; the deeper the layers, the higher level is the information they provide. These results were obtained after a set of only 38 training CTAs, after which the algorithms were not only able to correctly detect the global volume, but also to automatically segment the lumen and the thrombus.

The number of required training CTAs was low compared with the literature. Indeed, Dai et al[12] reported the use of 208 training CTAs for 103 tested CTAs for deep learning for automated cerebral aneurysm detection, whereas Park et al[13] reported a training set of 611 CTAs for a test set of 115 CTAs. As for deep learning approaches for AAAs, Jiang et al[14] reported the use of 6 sets of patient data from 6 different patients randomly selected as testing data among 55 sets of interpolated patient data as collections from 20 patients, whereas Hahn et al[15] reported a training set of 101 CTAs for a test set of 33 fully segmented CTAs. This difference is explained by the fact that the PRAEVAorta software mixes artificial intelligence and data modeling to reduce the training time and the amount of training data needed. This factor represents a great advantage in comparison with other solutions because it requires minimal labeling time (when manual segmentation is extremely time consuming) to extend the solution to other arterial segments, and the software is not limited anymore to the abdominal aorta as it is actually able to segment the whole aorta, the iliac, the femoropopliteal, renal, visceral and carotid arteries and it is now learning to segment the calcifications (Supplementary Fig 5, online only).

Lareyre et al[16] recently proposed a fully automated pipeline of their own to detect the aortic lumen and to characterize AAAs, including the presence of intraluminal thrombus and calcifications. In this work, they used a feature-based approach, in which the boundary propagation and active contour methods are used to segment the lumen. The method was tested on a set of 40 CTAs and also demonstrated a good correlation with results obtained from manual segmentation by a human expert, with a computational time of less than 1 minute per patient.[17] However, the analysis was performed on only 620 selected slices for the aortic lumen and 525 slices for the thrombus, whereas our study included all the 13,465 slices extracted from the 100 included CTAs and analyzed at once the global volume, the thrombus and the lumen in each slice with a computational time ranging from 27 seconds to 4 minutes. Their results were inferior to ours in terms of VS (0.96 ± 0.04), sensitivity (0.90 ± 0.06), JAC (0.87 ± 0.07), and DSC (0.93 ± 0.04) for global volume. The DSC for the detection of thrombus was slightly higher (0.88 ± 0.12), but was only

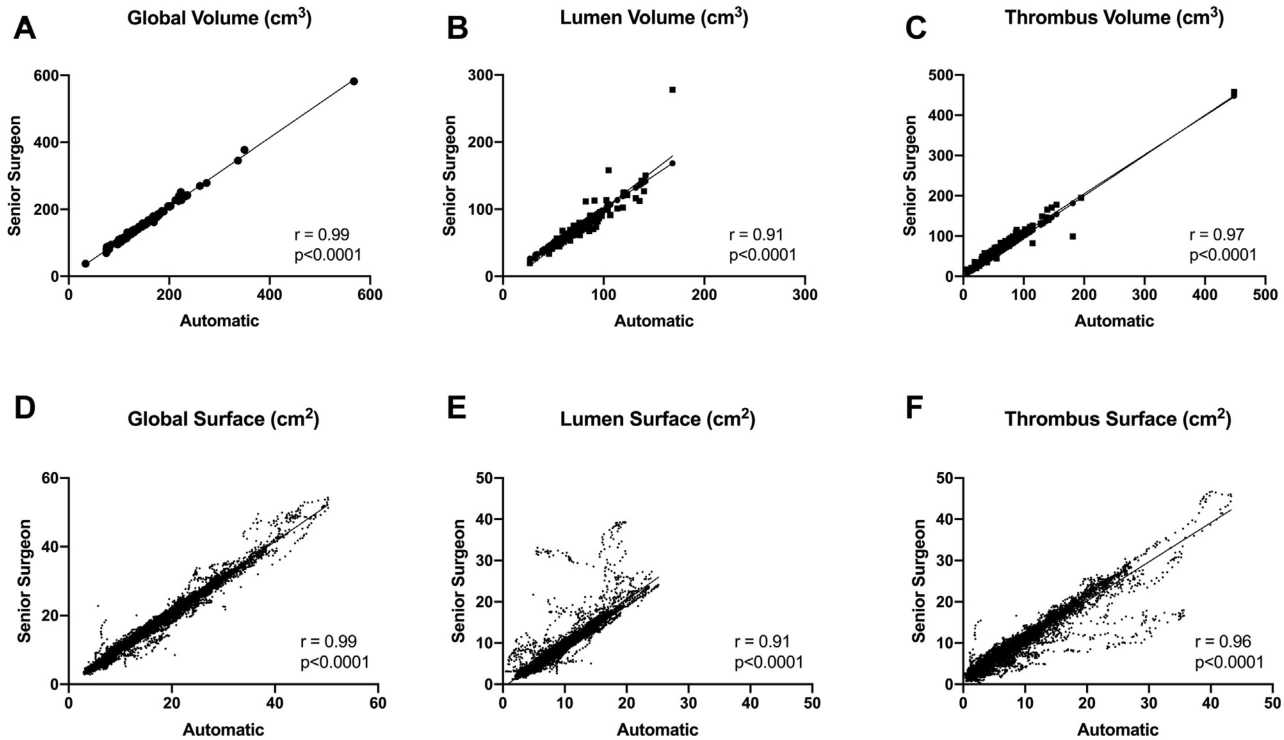


Fig 2. Correlation between the fully automatic segmentation and the manually corrected segmentation by an expert senior surgeon. **A,** Correlation of the aortic lumen's volume segmentation (n = 100). **B,** Correlation of the thrombus' volume segmentation (n = 100). **C,** Correlation of the global volume's segmentation (n = 100). **D,** Correlation of the aortic lumen's surface segmentation (n = 13,541). **E,** Correlation of the thrombus' surface segmentation (n = 13,541). **F,** Correlation of the global surface's segmentation (n = 13,541).

provided for a limited number of slices. Moreover, in the 40 test cases, 6 cases had one to three slices with false positives and 6 cases included false-negatives parts (aorta portion or a renal arteria departure undetected).

The intraluminal thrombus is a factor contributing to AAA development, and its increase correlates with the aneurysm sac and rupture.[18] A precise quantitative analysis of the thrombus would be useful to better assess the risk of AAA rupture. Thrombus segmentation is a challenging task because its borders are irregular and not well-defined because of the presence of adjacent structures with similar intensity values and regions of low boundary contrast. Several semiautomatic approaches have been proposed for thrombus segmentation.[19] López-Linares et al[8] also developed fully automatic approaches for thrombus detection based on a deep convolutional neural network. The pipeline was trained, validated, and tested in 13 postoperative CTA images of patients with AAA, and segmentation quality was evaluated by comparing results obtained from the automatic method with the manually delimited volume. It provided a DSC of 0.82 for postoperative thrombus segmentation without the need for human intervention in most common cases. As such, our DSC of 0.81 (vs senior surgeon) and 0.82 (vs junior surgeon) without any human intervention is promising. Nonetheless, a reversed dispersion of the thrombus and the lumen variables is observed in Fig 2, *E* and *F*, which is explained by the fact that the overall correlation is good and that when the lumen is not detected, it is usually counted as thrombus.

The artificial intelligence-derived software used in our study may improve image segmentation and analysis by allowing investigators to more easily detect the aneurysm, characterize its anatomic characteristics (including the presence of intraluminal thrombus) and automatically calculate the diameters, lengths, and volumes of the aneurysm. It has the advantage of reducing segmentation time. Hence, the semiautomatic method used in this work, based on contouring thrombus while the lumen is automatically segmented, lasted an average of 35 minutes, whereas other semiautomatic methods described in the literature report a segmentation time around 10 minutes,[19] and the whole image segmentation was performed in an average of 100 seconds using the PRAEVAorta software.

It also decreased user interaction, whereas semiautomatic methods remain operator dependent, which may improve image interpretation. Indeed, after careful review of the areas of discrepancy between the results obtained with the fully automatic method and the manually corrected results, it seems that the

Table I. Summarized results of the evaluation of the segmentation methodology for the summation of the aortic lumen and thrombus volumes on a dataset of 100 computed tomography (CT) scans (13,465 slices) obtained from patients with infrarenal abdominal aortic aneurysms (AAAs)

Variables	Mean ± SD	Minimum-Maximum	Median	Interquartile range
Surface from fully automatic method, cm^2	150.3 ± 67.1	33.4-567.7	134.8	111.4-169.9
Surface from manual method/senior surgeon, cm^2	156.1 ± 69.5	37.6-582.2	140.2	114.7-174.6
Surface from manual method/junior surgeon, cm^2	157.4 ± 70.1	37.7-577.1	141.8	114.4-177.4
Volume from fully automatic method, cm^3	150,275 ± 67,131	33,432-567,737	134,762	111,440-169,927
Volume from manual method/senior surgeon, cm^3	156,106 ± 69,484	37,571-582,150	140,157	114,679-174,584
Volume from manual method/junior surgeon, cm^3	157,400 ± 70,113	37,667-577,094	141,773	114,374-178,128
Computational analysis time/fully automatic method, s	90 ± 35	27-235	93	63-106
Computational analysis time/senior surgeon, s	1952 ± 1052	600-4680	1650	1080-2580
Computational analysis time/junior surgeon, s	2156 ± 1042	300-4800	1920	1380-2715
No. of slices/patient	135 ± 55	18-312	145	93-172
Automatic vs senior surgeon				
DSC	0.95 ± 0.01[a]	0.92-0.98	0.96	0.95-0.96
JAC	0.91 ± 0.02	0.85-0.96	0.92	0.90-0.93
Sensitivity	0.94 ± 0.02	0.86-0.97	0.94	0.92-0.95
Specificity	0.97 ± 0.01	0.92-0.99	0.97	0.96-0.98
Volume similarity	0.98 ± 0.01	0.93-1.00	0.98	0.97-0.99
HD/slice, mm	4.14 ± 6.20	0.70-63.18	2.34	1.83-3.46
Automatic vs junior surgeon				
DSC	0.95 ± 0.01[a]	0.92-0.98	0.96	0.94-0.96
JAC	0.91 ± 0.03	0.85-0.96	0.92	0.89-0.93
Sensitivity	0.93 ± 0.02	0.86-0.98	0.94	0.92-0.95
Specificity	0.97 ± 0.02	0.91-0.99	0.97	0.97-0.98
Volume similarity	0.98 ± 0.02	0.93-1.00	0.98	0.97-0.99
HD/slice, mm	4.14 ± 6.24	0.58-63.69	2.32	1.79-3.58
Senior surgeon vs junior surgeon				
DSC	0.98 ± 0.01[b]	0.92-1.00	0.98	0.97-0.99
JAC	0.96 ± 0.03	0.85-1.00	0.97	0.95-0.98
Volume similarity	0.99 ± 0.01	0.94-1.00	0.99	0.99-1.00
HD/slice, mm	1.84 ± 2.24	1.00-52.82	1.71	0.73-2.56

DSC, Dice similarity coefficient; *HD*, Hausdorff distance; *JAC*, Jaccard index; *SD*, standard deviation.
[a]DSC/slice: 0.95 ± 0.05.
[b]DSC/slice: 0.98 ± 0.06.

results of the software surpassed the interpretation of the expert surgeon in some cases. So even though the semiautomatic manual segmentations demonstrated low interobserver variability, as already demonstrated in the literature,[20] it can still lead to human error.

In addition, after careful reconsideration of the CTA segmentations, it seems that the lumen/thrombus distinction sometimes presents significant errors in the manually corrected segmentations owing to the use of semiautomatic segmentation with a lumen volume that is generally lower than that detected in automatic segmentation. Thus, this zone not detected as lumen is then detected as thrombus, the volume of which is wrongly increased, which tends to show a lower DSC despite a correct detection in automatic segmentation.

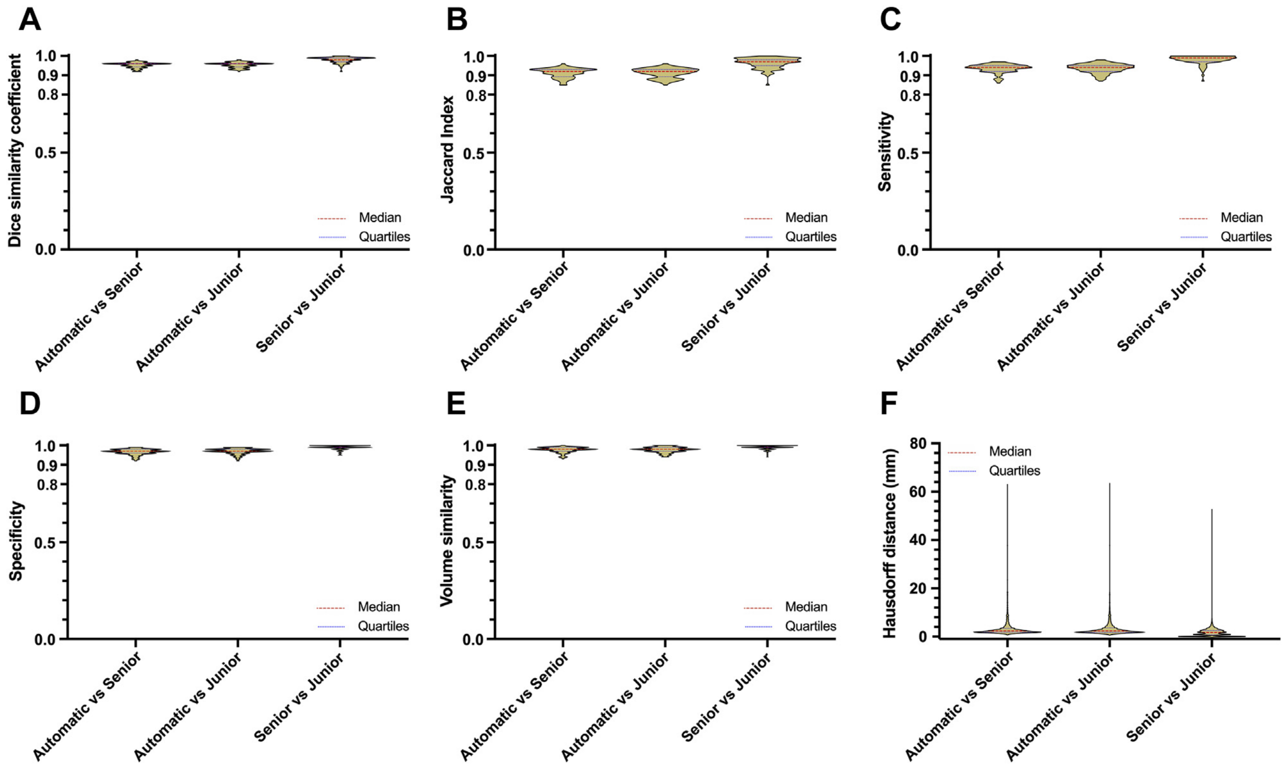


Fig 3. Summarized results of the quantitative evaluation of the fully automatic segmentation method (*automatic*) compared with the manually corrected segmentations provided by a senior surgeon (*senior*) and a junior surgeon (*junior*). **A,** Dice similarity coefficient (DSC), **(B)** Jaccard index (JAC), **(C)** sensitivity, **(D)** specificity, **E,** volume similarity, and **F,** Hausdorff distance (HD).

As for the extremes of HD, these cases are mainly where the upper limit of the segmentation was fixed just at the level of the renal arteries, with an angulated neck. Current algorithms then included the renal artery's ostium in the segmentation, whereas the manually corrected segmentations removed them, thus generating significant errors on only a few slices.

We found no evidence of a relationship between false positives or false negatives and slice thickness, because it induces the same difficulties for manual segmentation. Nonetheless, the algorithms still require additional machine learning to correctly identify the limits of the AAA when it is close to the vena cava and the duodenum, or when it presents with meaningful wall inflammation, noise, or artifacts. Indeed, the threshold value appears to be of utmost importance, it is automatically attributed by the software's algorithm and designed to limit propagation in other tissues touching the aorta (false positive) while promoting detection in areas with low contrast product often related to bad acquisition timing or artifacts secondary to metallic implants (false negative). False negatives were found for poorly contrasted CTAs, where the lumen is weakly injected, leading to a low-intensity level (<200 HU). Thereby, the quality of contrast liquid injection remains a limiting factor, and some CTAs had an intensity barely above 100 HU on certain segments of the aorta, especially the distal portion of the abdominal aorta downstream from the turbulence linked to AAA, when the level expected after contrast medium injection is 200 HU. As such, the lumen may not be detected by the algorithms and because the algorithms analyzing the thrombus need the lumen identification, the thrombus may as well be undetected. Additional post-treatment operations based on intensity similarity and morphologic operations have been implemented in the latest version of the algorithms and seem to now limit aberrant propagation to adjacent structures. In addition, 19 CTAs included in the study had mean lumen intensities of less than 200 HU and were still accurately treated, showing good robustness of the algorithms. Although the algorithms still perform well on poorly contrasted CTAs, particular care has to be taken to decrease the dependency of the results to contrast level. As such, it would be particularly interesting to be able to treat noncontrast CTs in the future.

Importantly, the 100 CTAs included in our study came from four different scan manufacturers, 53 from General Electric scanners, 18 from Siemens and Philips scanners, 10 from Toshiba, and 1 from an unknown manufacturer. We observed no performance differences between

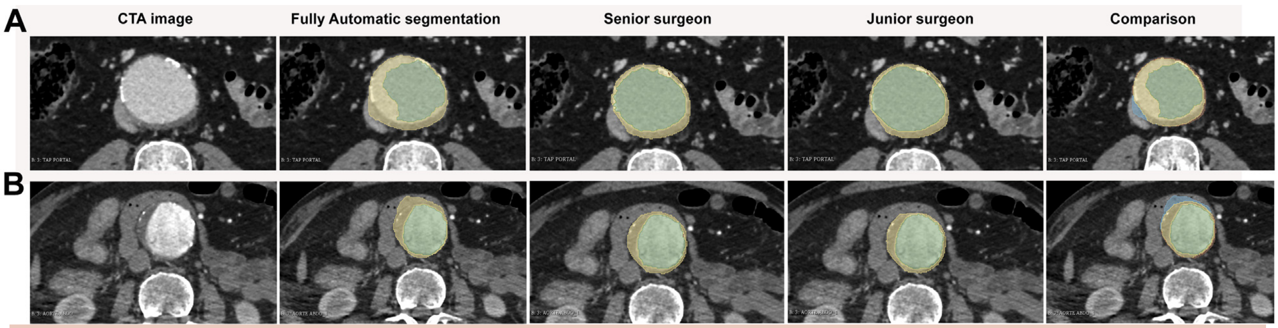


Fig 4. Representative images of segmentation areas with false positives. Computed tomography angiography (*CTA*) cross-sections from patients with an infrarenal abdominal aortic aneurysm (AAA) analyzed by fully automatic segmentation, manually corrected segmentation by the senior surgeon and the junior surgeon, showing the aortic lumen (*red*) and the intraluminal thrombus (*green*). A comparison of the results obtained with the fully automatic segmentation and the manually corrected segmentation by the senior surgeon are also displayed, showing the lumen common to manual segmentation and automatic segmentation (*green*), the thrombus common to manual segmentation and automatic segmentation (*yellow*), the false negatives (*red*), and the false positives (*blue*). **A,** The vena cava seems to be partly in blue owing to a false recognition by the automatic software as part of the AAA because of a close proximity of the shades of grey between the vena cava, the thrombus and the lumen caused by improperly applied contrast liquid close to a portal phase. **B,** The duodenum appears partly in blue owing to a false recognition by the automatic software as part of the AAA because of a close proximity of the shades of grey between the duodenum and the thrombus.

Table II. Summarized results of the evaluation of the maximum aortic transverse diameter on a dataset of 100 computed tomography (CT) scans obtained from patients with infrarenal abdominal aortic aneurysms (AAAs)

Variables	Mean ± SD	Minimum-Maximum	Median	Interquartile range
Maximum aortic transverse diameter from fully automatic method, mm	54.9 ± 8.8	32.8-85.5	53.1	50.2-57.4
Maximum aortic transverse diameter/senior surgeon, mm	54.3 ± 8.5	30.0-85.0	53.0	50.0-56.0
Maximum aortic transverse diameter/junior surgeon, mm	55.1 ± 8.0	37.9-82.5	53.9	50.6-57.8
Δ Fully automatic − senior surgeon,[a] mm	2.3 ± 2.1	0.0-13.6	1.8	0.7-3.5
Δ Fully automatic − junior surgeon,[a] mm	3.4 ± 4.2	0.0-24.9	2.0	0.7-4.9
Δ Senior surgeon − junior surgeon,[b] mm	2.8 ± 3.8	0.0-27.7	1.7	0.8-3.1

SD, Standard deviation.
[a]Absolute number comparison between the maximum aortic transverse diameter obtained with the fully automatic method and the surgeon.
[b]Absolute number comparison between the maximum aortic transverse diameter obtained by the two surgeons.

manufacturers showing the applicability of PRAEVAorta to any type of scanner.

In routine clinical practice, the size of an AAA is determined by manual measurement of the maximal aortic diameter, which is time consuming and prone to high interobserver variability. Very few articles on deep learning approaches report the evaluation of such diameters in addition to the volumes. The 2.3-mm mean discrepancy provided by our algorithm is well within the 5-mm gradations generally observed between radiologists and close to the results of Lu et al showing a 2.02 mm average discrepancy despite lower DSC (0.91), sensitivity (0.91), and specificity (0.95) for volume segmentation.[21] Their algorithm was also used on noncontrast CTs, but was unable to detect the difference between lumen and thrombus on CTAs. Like us, they used aorta contour fitting for the estimation of the largest cross-sectional diameter and assigned it to the long axis of the ellipses.

Hence, these advances may help to better evaluate the prognosis of patients by precisely characterizing the AAA while avoiding the risk of human error, and even predict the risk of AAA growth and rupture.[22] This standardized characterization of AAAs could be useful in clinical practice to homogenize and facilitate the therapeutic indication and offers interesting perspectives to easily and quickly characterize a high quantity of CT scans, which could be useful for applications in clinical research and could potentially decrease the costs.[23]

Limitations. Our analysis was limited to strict infrarenal AAAs because the precise characterization of patients with complex aneurysms, including the

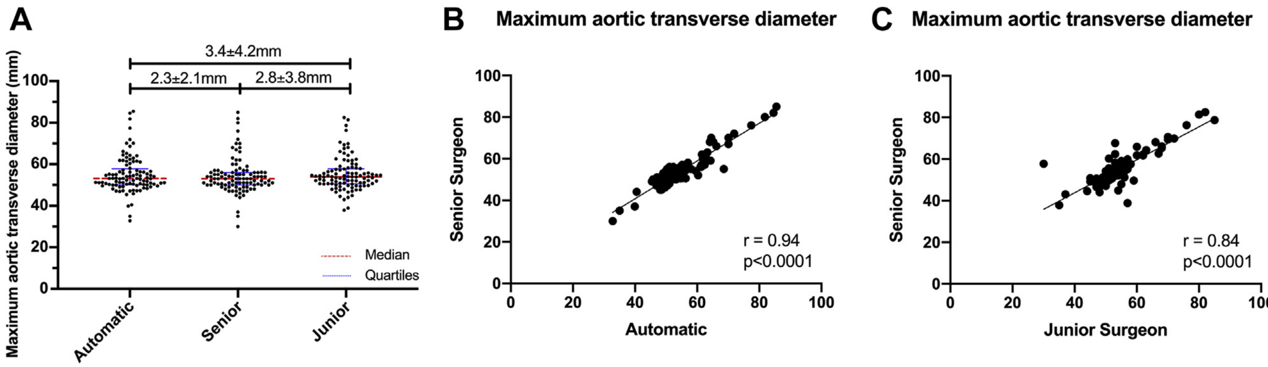


Fig 5. Analysis of the maximum aortic transverse diameter. **A,** Results as measured by the fully automatic segmentation method (*automatic*) compared with the manually corrected segmentations provided by a senior surgeon (*senior*) and a junior surgeon (*junior*), the mean diameter ± standard deviation difference between the techniques of measurement are provided above the bars, and the median (range) are provided below the bars. **B,** Correlation between the fully automatic method and the senior surgeon (n = 100). **C,** Correlation between the senior and the junior surgeon (n = 100). r, Pearson's coefficient correlation.

pararenal and visceral segments, still requires future validation. In its current version, the PRAEVAorta software can differentiate between juxtarenal/suprarenal aneurysms (Supplementary Fig 5, online only) and purely infrarenal ones, but improvements remain necessary to hopefully include automatic naming of the target arteries, geometrical analysis, and even postoperative analysis.

Our results establish the feasibility and ease of use of this software in more than 100 CTAs carried out as part of the preoperative assessment in different radiology centers in our region. The generalizability of those results and the applicability to postoperative surveillance may need multicenter registries taking into account the broad spectrum of patient demographics and anatomies before these techniques and approaches can be used in daily clinical practice; we are currently working on a collaboration with other European and North American centers to provide such data. The ultimate aim of this study is to detect for rapid aneurysm evolution during preoperative follow-up and to implement this technique to the postoperative follow-up, so that endoleaks can be detected easily and treated as soon as they are responsible for aneurysm sac evolution.

CONCLUSIONS

Fully automatic volume segmentation of baseline CTAs of AAAs demonstrated high accuracy and high correlation with the results of manually corrected segmentation performed by a senior and a junior vascular surgeon regarding the aortic lumen, thrombus, global volume, and diameters, with important time gain. This fully automatic method brings a standardized characterization of AAA and could have strong applications in daily clinical practice and clinical research.

AUTHOR CONTRIBUTIONS

Conception and design: CC, ED
Analysis and interpretation: CC, XB
Data collection: BS, AV
Writing the article: CC, ED
Critical revision of the article: CC, BS, AV, XB, ED
Final approval of the article: CC, BS, AV, XB, ED
Statistical analysis: CC
Obtained funding: Not applicable
Overall responsibility: ED

REFERENCES

1. Reimerink JJ, van der Laan MJ, Koelemay MJ, Balm R, Legemate DA. Systematic review and meta-analysis of population-based mortality from ruptured abdominal aortic aneurysm. Br J Surg 2013;100:1405-13.
2. Parkinson F, Ferguson S, Lewis P, Williams IM, Twine CP. Rupture rates of untreated large abdominal aortic aneurysms in patients unfit for elective repair. J Vasc Surg 2015;61:1606-12.
3. Parker MV, O'Donnell SD, Chang AS, Johnson CA, Gillespie DL, Goff JM, et al. What imaging studies are necessary for abdominal aortic endograft sizing? A prospective blinded study using conventional computed tomography, aortography, and three-dimensional computed tomography. J Vasc Surg 2005;41:199-205.
4. Wanhainen A, Verzini F, Herzeele IV, Allaire E, Bown M, Cohnert T, et al. Editor's Choice - European Society for Vascular Surgery (ESVS) 2019 Clinical Practice Guidelines on the management of abdominal aorto-iliac artery aneurysms. Eur J Vasc Endovasc Surg 2019;57:8-93.
5. Mora CE, Marcus CD, Barbe CM, Ecarnot FB, Long AL. Maximum diameter of native abdominal aortic aneurysm measured by angio-computed tomography: reproducibility and lack of consensus impacts on clinical decisions. Aorta Stamford Conn 2015;3:47-55.
6. Lindquist Liljeqvist M, Hultgren R, Gasser TC, Roy J. Volume growth of abdominal aortic aneurysms correlates with baseline volume and increasing finite element analysis-derived rupture risk. J Vasc Surg 2016;63:1434-1442.e3.
7. Joldes GR, Miller K, Wittek A, Forsythe RO, Newby DE, Doyle BJ. BioPARR: a software system for estimating the rupture potential index for abdominal aortic aneurysms. Sci Rep 2017;7:1-15.
8. López-Linares K, Aranjuelo N, Kabongo L, Maclair G, Lete N, Ceresa M, et al. Fully automatic detection and segmentation of abdominal aortic thrombus in post-operative CTA images using Deep Convolutional Neural Networks. Med Image Anal 2018;46:202-14.

9. Maiora J, Papakostas GA, Kaburlasos VG, Grana M. A proposal of texture features for interactive CTA segmentation by Active Learning. Stud Health Technol Inform 2014;207:311-20.
10. Taha AA, Hanbury A. Metrics for evaluating 3D medical image segmentation: analysis, selection, and tool. BMC Med Imaging 2015;15:29.
11. Rote G. Computing the minimum Hausdorff distance between two point sets on a line under translation. Inf Process Lett 1991;38: 123-7.
12. Dai X, Huang L, Qian Y, Xia S, Chong W, Liu J, et al. Deep learning for automated cerebral aneurysm detection on computed tomography images. Int J Comput Assist Radiol Surg 2020;15:715-23.
13. Park A, Chute C, Rajpurkar P, Lou J, Ball RL, Shpanskaya K, et al. Deep learning-assisted diagnosis of cerebral aneurysms using the HeadXNet model. JAMA Netw Open 2019;2:e195600.
14. Jiang Z, Do HN, Choi J, Lee W, Baek S. A deep learning approach to predict abdominal aortic aneurysm expansion using longitudinal data. Available at: www.frontiersin.org/articles/10.3389/fphy.2019.00235/full. Accessed March 6, 2021.
15. Hahn S, Perry M, Morris CS, Wshah S, Bertges DJ. Machine deep learning accurately detects endoleak after endovascular abdominal aortic aneurysm repair. JVS Vasc Sci 2020;1:5-12.
16. Lareyre F, Adam C, Carrier M, Dommerc C, Mialhe C, Raffort J. A fully automated pipeline for mining abdominal aortic aneurysm using image segmentation. Sci Rep 2019;9:13750.
17. Swedenborg J, Eriksson P. The intraluminal thrombus as a source of proteolytic activity. Ann NY Acad Sci 2006;1085:133-8.
18. de Bruijne M, van Ginneken B, Viergever MA, Niessen WJ. Adapting active shape models for 3D segmentation of tubular structures in medical images. In: Taylor C, Noble JA, editors. Information processing in medical imaging. Berlin, Heidelberg: Springer; 2003. p. 136-47. (Lecture notes in computer science).
19. Gao X, Boccalini S, Kitslaar PH, Budde RPJ, Tu S, Lelieveldt BPF, et al. A novel software tool for semi-automatic quantification of thoracic aorta dilatation on baseline and follow-up computed tomography angiography. Int J Cardiovasc Imaging 2019;35:711-23.
20. van Prehn J, van der Wal MBA, Vincken K, Bartels LW, Moll FL, van Herwaarden JA. Intra- and interobserver variability of aortic aneurysm volume measurement with fast CTA postprocessing software. J Endovasc Ther 2008;15:504-10.
21. Lu J-T, Brooks R, Hahn S, Chen J, Buch V, Kotecha G, et al. Deep AAA: clinically applicable and generalizable detection of abdominal aortic aneurysm using deep learning. Available at: http://arxiv.org/abs/1907.02567. Accessed March 6, 2021.
22. Lee R, Jarchi D, Perera R, Jones A, Cassimjee I, Handa A, et al. Applied machine learning for the prediction of growth of abdominal aortic aneurysm in humans. EJVES Short Rep 2018;39:24-8.
23. Dey D, Slomka PJ, Leeson P, Comaniciu D, Shrestha S, Sengupta PP, et al. Artificial intelligence in cardiovascular imaging: JACC State-of-the-Art Review. J Am Coll Cardiol 2019;73:1317-35.

Submitted Jul 21, 2020; accepted Nov 12, 2020.

Additional material for this article may be found online at www.jvascsurg.org.

Supplementary Table I (online only). Summarized results of the evaluation of the segmentation methodology for the aortic lumen on a dataset of 100 computed tomography (CT) scans (13,465 slices) obtained from patients with infrarenal abdominal aortic aneurysms (AAAs)

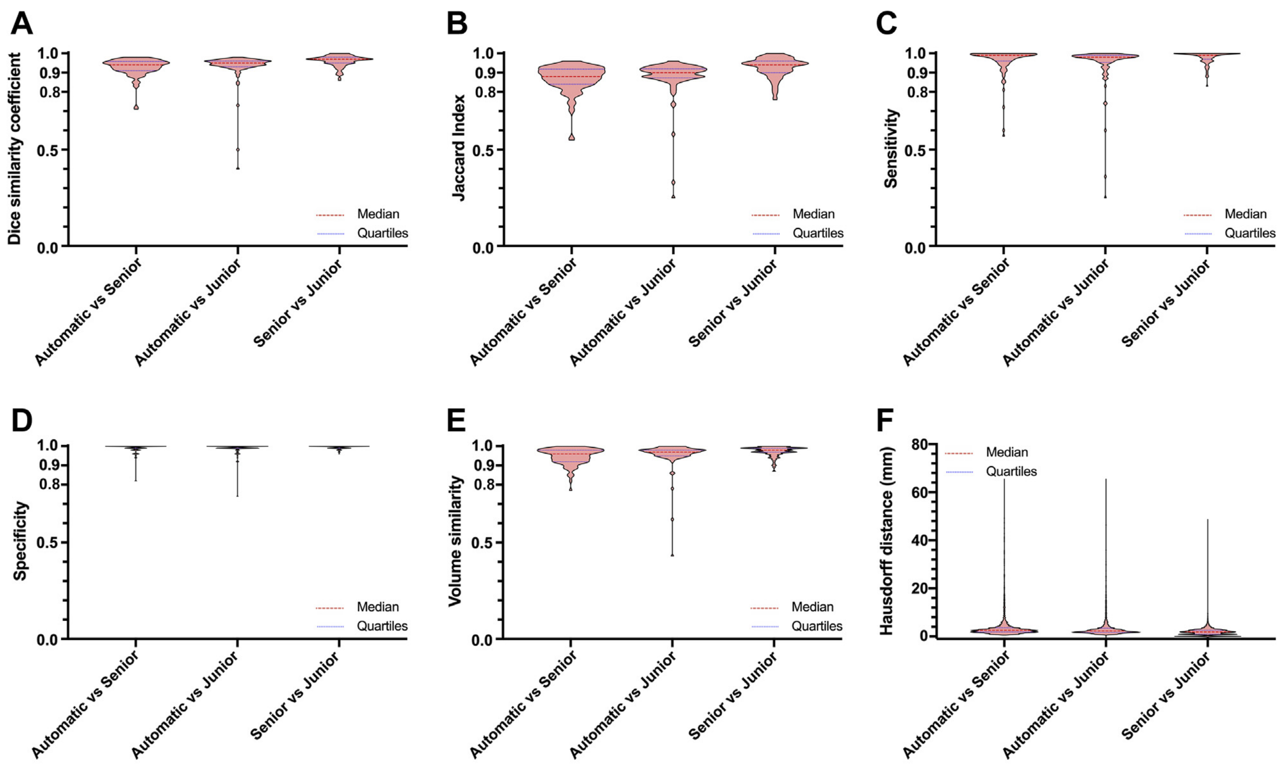
Variables	Mean ± SD	Minimum-Maximum	Median	Interquartile range
Surface from fully automatic method, cm^2	78.5 ± 26.8	26.7-168.4	76.0	58.8-93.7
Surface from manual method/senior surgeon, cm^2	75.5 ± 34.1	19.9-278.2	70.7	53.2-90.1
Surface from manual method/junior surgeon, cm^2	78.0 ± 35.1	24.2-292.6	73.2	55.6-90.8
Volume from fully automatic method, cm^3	78,359 ± 26,754	26,721-168,430	75,957	58,432-93,954
Volume from manual method/senior surgeon, cm^3	75,516 ± 34,052	19,924-278,164	70,679	52,930-90,166
Volume from manual method/junior surgeon, cm^3	77,995 ± 35,097	24,150-292,637	73,195	54,914-90,898
Automatic vs senior surgeon				
DSC	0.93 ± 0.05	0.71-0.98	0.94	0.91-0.95
JAC	0.87 ± 0.07	0.55-0.96	0.88	0.84-0.91
Sensitivity	0.96 ± 0.07	0.57-1.00	0.99	0.96-1.00
Specificity	0.99 ± 0.02	0.82-1.00	1.00	0.99-1.00
Volume similarity	0.95 ± 0.04	0.77-1.00	0.96	0.92-0.98
HD/slice, mm	4.61 ± 7.26	0.59-65.71	2.56	1.83-3.68
Automatic vs junior surgeon				
DSC	0.93 ± 0.08	0.40-0.98	0.95	0.93-0.96
JAC	0.88 ± 0.10	0.25-0.96	0.90	0.88-0.92
Sensitivity	0.95 ± 0.11	0.25-1.00	0.98	0.96-0.99
Specificity	0.99 ± 0.03	0.74-1.00	1.00	0.99-1.00
Volume similarity	0.96 ± 0.07	0.43-1.00	0.97	0.95-0.98
HD/slice, mm	4.36 ± 7.27	1.00-65.71	2.27	1.66-3.42
Senior surgeon vs junior surgeon				
DSC	0.96 ± 0.03	0.86-1.00	0.97	0.95-0.98
JAC	0.93 ± 0.05	0.76-1.00	0.94	0.90-0.96
Volume similarity	0.97 ± 0.03	0.87-1.00	0.98	0.97-0.99
HD/slice, mm	1.97 ± 1.80	1.00-48.91	1.83	0.93-2.56

DSC, Dice similarity coefficient; *HD*, Hausdorff distance; *JAC*, Jaccard index; *SD*, standard deviation.

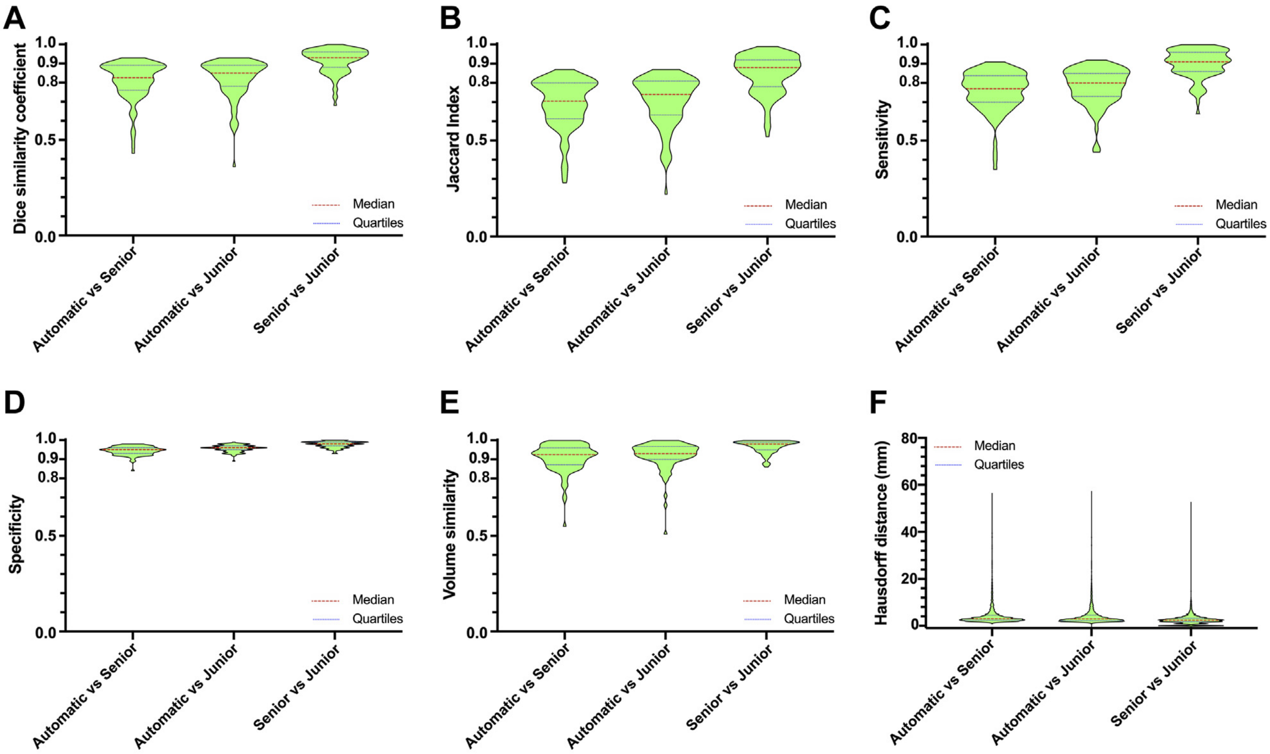
Supplementary Table II (online only). Summarized results of the evaluation of the segmentation methodology for the thrombus on a dataset of 100 computed tomography (CT) scans (13,465 slices) obtained from patients with infrarenal abdominal aortic aneurysms (AAAs)

Variables	Mean ± SD	Minimum-Maximum	Median	Interquartile range
Surface from fully automatic method, cm^2	71.7 ± 52.9	6.7-448.2	62.1	40.5-82.6
Surface from manual method/senior surgeon, cm^2	80.5 ± 52.8	17.6-458.0	72.2	48.8-95.0
Surface from manual method/junior surgeon, cm^2	78.0 ± 52.0	13.5-450.8	72.1	46.4-88.2
Volume from fully automatic method, cm^3	71,688 ± 52,935	6711-448,158	62,108	39,936-83,778
Volume from manual method/senior surgeon, cm^3	80,453 ± 52,763	17,648-457,978	72,208	48,580-95,193
Volume from manual method/junior surgeon, cm^3	78,000 ± 51991	13,518-450,846	72,146	46,283-90,726
Automatic vs senior surgeon				
DSC	0.81 ± 0.10	0.43-0.93	0.83	0.76-0.89
JAC	0.69 ± 0.13	0.28-0.87	0.71	0.62-0.80
Sensitivity	0.76 ± 0.10	0.35-0.91	0.77	0.70-0.83
Specificity	0.94 ± 0.02	0.84-0.98	0.95	0.93-0.96
Volume similarity	0.91 ± 0.07	0.55-1.00	0.93	0.88-0.96
HD/slice, mm	4.76 ± 5.55	0.85-56.58	3.02	2.37-4.45
Automatic vs junior surgeon				
DSC	0.82 ± 0.10	0.36-0.93	0.85	0.78-0.89
JAC	0.70 ± 0.13	0.52-0.87	0.74	0.64-0.81
Sensitivity	0.78 ± 0.10	0.44-0.92	0.80	0.73-0.85
Specificity	0.96 ± 0.02	0.89-0.99	0.96	0.95-0.97
Volume similarity	0.92 ± 0.07	0.51-1.00	0.93	0.90-0.96
HD/slice, mm	4.71 ± 5.66	1.00-57.39	2.93	2.19-4.51
Senior surgeon vs junior surgeon				
DSC	0.91 ± 0.06	0.68-1.00	0.93	0.88-0.96
JAC	0.85 ± 0.10	0.52-0.99	0.88	0.78-0.92
Volume similarity	0.97 ± 0.03	0.86-1.00	0.98	0.95-0.99
HD/slice, mm	2.71 ± 2.52	1.00-52.82	2.39	1.71-3.20

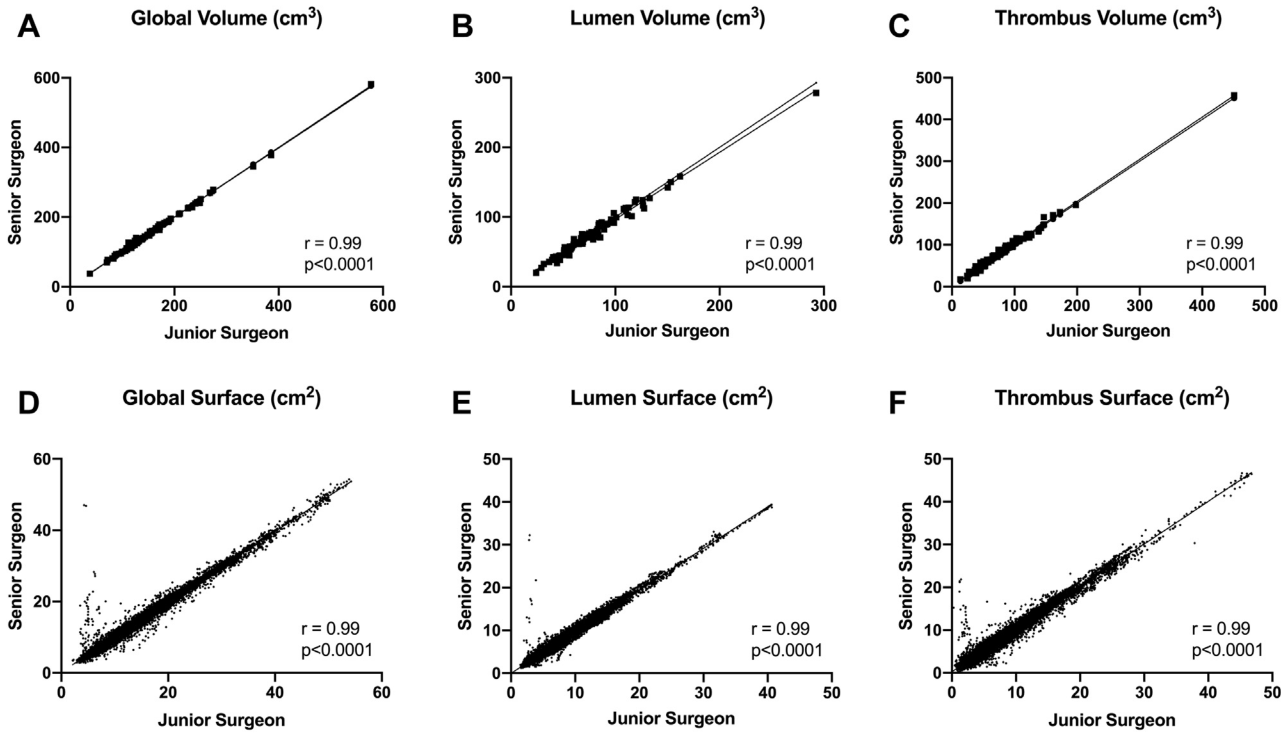
DSC, Dice similarity coefficient; *HD*, Hausdorff distance; *JAC*, Jaccard index; *SD*, standard deviation.



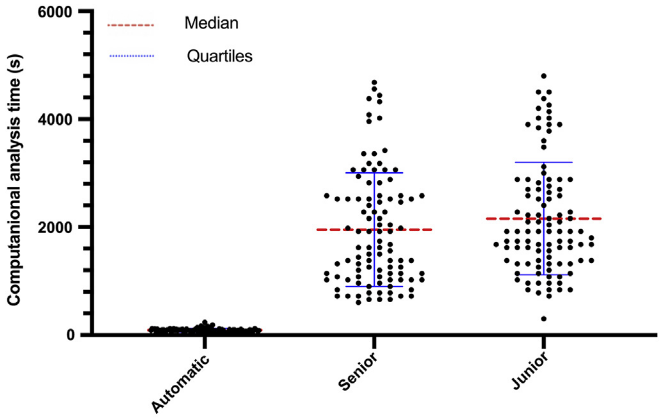
Supplementary Fig 1 (online only). Summarized results of the quantitative evaluation of the lumen using the fully automatic segmentation method (*automatic*) compared with the manually corrected segmentations provided by a senior surgeon (*senior*) and a junior surgeon (*junior*). **A,** Dice similarity coefficient (DSC), **(B)** Jaccard index (JAC), **(C)** sensitivity, **(D)** specificity, **(E)** volume similarity, and **(F)** Hausdorff distance.



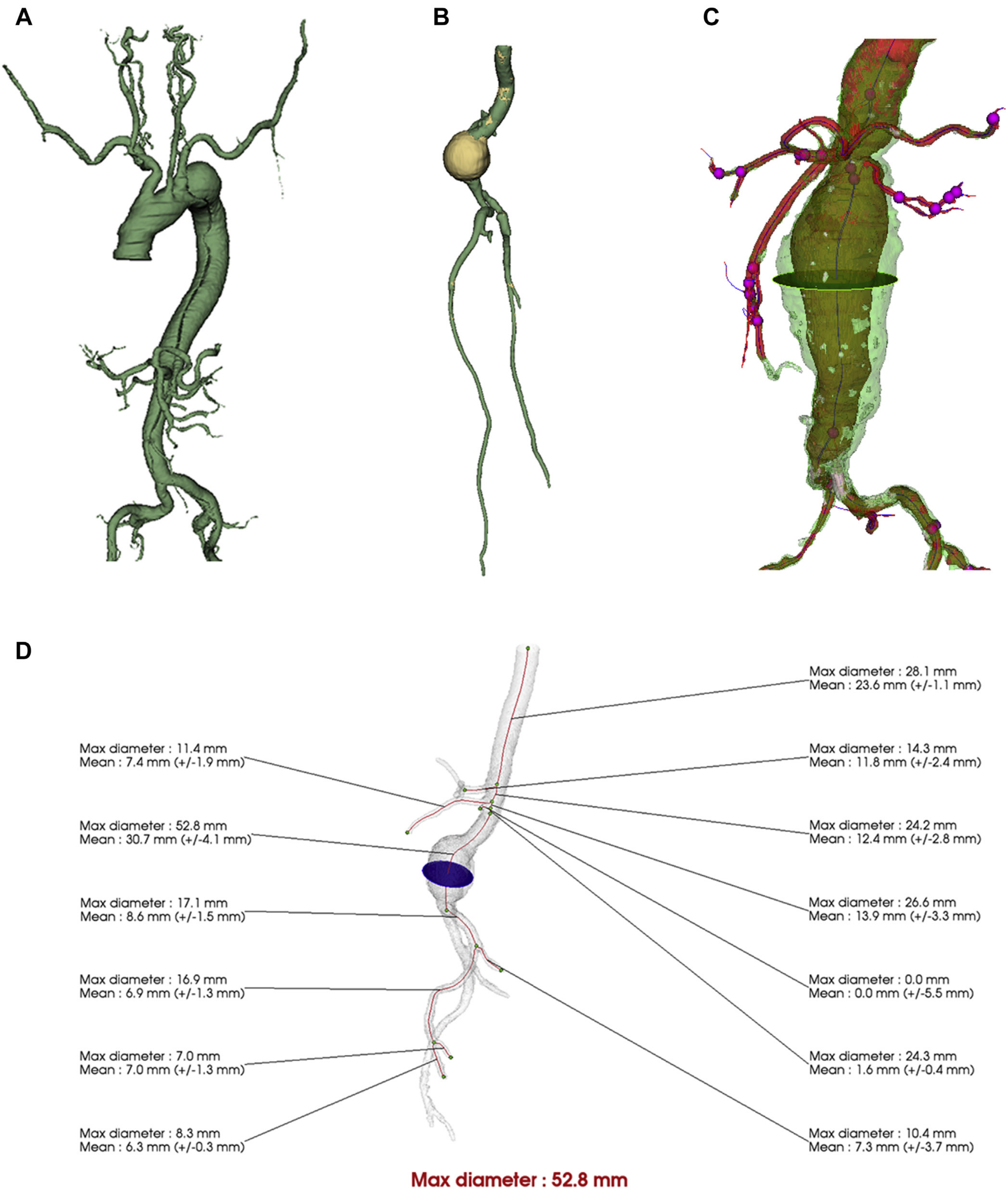
Supplementary Fig 2 (online only). Summarized results of the quantitative evaluation of the thrombus using the fully automatic segmentation method (*automatic*) compared with the manually corrected segmentations provided by a senior surgeon (*senior*) and a junior surgeon (*junior*). **A,** Dice similarity coefficient (DSC), **(B)** Jaccard index (JAC), **(C)** sensitivity, **(D)** specificity, **(E)** volume similarity, and **(F)** Hausdorff distance.



Supplementary Fig 3 (online only). Correlation between the manually corrected segmentation by an expert senior surgeon and a junior surgeon. **A,** Correlation of the global volume's segmentation (n = 100), **(B)** correlation of the aortic lumen's volume segmentation (n = 100), **(C)** correlation of the thrombus' volume segmentation (n = 100), **(D)** correlation of the global surface's segmentation (n = 13,541), (E) correlation of the aortic lumen's surface segmentation (n = 13,541), **(F)** correlation of the thrombus' surface segmentation (n = 13,541). r, Pearson's coefficient correlation.



Supplementary Fig 4 (online only). Time required for the fully automatic segmentation method (*automatic*) compared with the manually corrected segmentations provided by a senior surgeon (*senior*) and a junior surgeon (*junior*).



Supplementary Fig 5 (online only). Illustrations of the applicability of the PRAEVAorta segmentation method to other arterial segments. **A,** Supra-aortic trunks and the entire aorto-iliac axis in a dissection. **B,** Femoropopliteal segments. **C,** Juxtarenal and pararenal aortic aneurysms with visceral and renal arteries. **D,** Preliminary export sheet for diameters in juxtarenal and pararenal aortic aneurysms.